Douwe Rijs

Robotics & Machine Learning Engineer · MSc Robotics, TU Delft

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Robotics MSc student at TU Delft with a Mechanical Engineering background and a strong focus on machine learning, perception and control systems. Hands-on R&D experience in computer vision, transformer models and enterprise software development at Allseas. Motivated to apply advanced robotics and ML to real-world industrial systems.

EXPERIENCE

Robotics & AI/ML Intern — Boeing Innovation Lab, Neu-Isenburg (Germany) — Sep – Dec 2026 (upcoming)

Mandatory MSc internship on autonomy and AI/ML applications for future flight operations.

Assistant R&D Engineer — Allseas, Delft — Feb 2025 – Aug 2026

- Developed an end-to-end computer-vision/VLM system for automated pipe-ID recognition from imagery of curved steel surfaces, increasing exact-ID accuracy from ~1% to ~80% — see Robotics & AI Projects.
- Built a network observability platform monitoring 100+ switches via SNMP, aggregating port-level telemetry (data rate, VLAN, link status) for real-time diagnostics.
- Developed and maintained production software in Python, Go and React/TypeScript within an enterprise R&D environment.
- Automated software versioning and releases through Azure DevOps CI/CD pipelines.

EDUCATION

MSc Robotics — Delft University of Technology — 2025 – present

Machine learning, deep reinforcement learning, machine perception (sensor fusion, Kalman filtering, LiDAR/radar/stereo vision), robot dynamics & control, motion planning (RRT, MPC).*

BSc Mechanical Engineering — Delft University of Technology — 2022 – Oct 2025

Thesis (8.0/10): chemotherapy-injection assisting device — medical-device regulations & CE certification. Minor in Computer Science: ML fundamentals, camera geometry & 3D reconstruction, algorithms & software engineering.

VWO (pre-university) — Haarlemmermeerlyceum, Hoofddorp — 2016 – 2022

ROBOTICS & AI PROJECTS

Robotics Engineer — Greenhouse Monitoring Robot & Digital Twin (MIRTE), TU Delft, team of 5

- Built an autonomous greenhouse robot in ROS 2 using Nav2 and SLAM Toolbox to map flower beds and stream plant, pest and climate data into a live digital twin.
- Developed the robot arm-control package and integrated SLAM, autonomous navigation, perception and state-machine components into a complete system; validated YOLOv8 detection and AprilTag localisation in simulation and on physical hardware.
- Managed the team's GitLab development workflow; delivered 7/8 client requirements within an 8-week development cycle.

Autonomy Engineer — Autonomous Micro Air Vehicle, TU Delft MAVLab

- Vision-only obstacle avoidance for a Parrot Bebop 2 flying autonomously in the CyberZoo using onboard compute only.
- Implemented border, gate and vertical-obstacle detection in C/C++ inside the Paparazzi autopilot; optimised the vision stack to 8–11 FPS on the drone's onboard ARM processor.

Computer Vision & VLM Engineer — Pipe-ID Recognition, Allseas R&D

- Developed an end-to-end CV/VLM pipeline reading stencilled & chalked pipe IDs from photos taken inside pipeline pipes, increasing exact-ID accuracy from ~1% to ~80%.
- Benchmarked four OCR architectures (Tesseract, EasyOCR, PaddleOCR, TrOCR), diagnosed failure modes and pivoted to a tuned VLM approach after classical OCR plateaued.
- Improved exact-match accuracy from 29% to 59% through YOLO oriented-bounding-box crop preprocessing alone — demonstrating that preprocessing outweighed model choice.
- Prototyped a stencil- and colour-agnostic ID locator using CRAFT text/linking scores with per-project arc-fitted text maps, generalising across pipelines with different layouts; evaluated 360° capture to remove blind-spot failures.

SKILLS

Programming	Python, Go, TypeScript/JavaScript, React, C++, MATLAB
Machine Learning	CNNs, RNNs, Transformers, VLMs, deep reinforcement learning, SVM, PCA, Hugging Face, YOLO
Robotics	ROS 2, SLAM, Nav2, sensor fusion, Kalman filtering, LiDAR, radar, stereo vision, pose estimation
Control & Planning	PID, MPC, impedance control, robot dynamics, RRT*, search algorithms
Tools & DevOps	Git, Azure DevOps, CI/CD, Docker, Linux, observability & monitoring
Languages	Dutch (native), English (fluent)